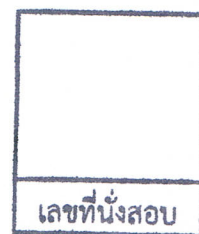


College of Industrial Technology
King Mongkut's University of Technology North Bangkok



Midterm Examination of semester 2

Year: 2014

Subject: 394172 Mathematics II

Section: 15-16

Date: 10 March 2015

Time: 10.00-12.00

Name: _____ ID: _____ Class: _____

Instructions

1. Cheating will result in failure of all classes registered for the current semester. Student who are caught cheating will also be denied registering for the following semester.
2. **No** documents are allowed to be taken out of the examination room.
3. Textbooks and dictionary are **NOT** allowed.
4. Calculators are allowed.
5. Electronic communication devices are **NOT** allowed in the examination room.
6. This examination has 8 pages (including this page), 11 questions do only 10 questions and a total score is 100.
7. Write all your answers on this examination sheet.

Question 1. Find the truth value of the compound statements

$[\neg p \rightarrow (q \vee r)] \rightarrow [\neg r \wedge (p \leftrightarrow \neg q)]$. The table with the different possible truth values has been provided. (Note $\neg$ means negation) (10 points)

p	q	r
T	T	T
T	T	F
T	F	T
T	F	F
F	T	T
F	T	F
F	F	T
F	F	F

Question 2. Let p , q and r be statements. The compound statement $p \rightarrow (q \wedge r)$ is false and $(p \vee q) \rightarrow r$ is true. Find the truth values of the following statement

$[p \leftrightarrow (q \leftrightarrow \neg r)] \wedge [(\neg p \rightarrow r) \vee q]$. (10 points)

Question 3. Given

p represents "If $a + c > b + c$, then $a > b$, where a, b, c are real numbers."

q represents "For all real numbers x , $|x| > 2$, if and only if $x \geq 2$." Which the following statements is not equivalent to $\neg p \vee q$.

1. $\neg p \wedge q$

2. $p \rightarrow q$

3. $\neg p \leftrightarrow q$

4. $\neg p \vee \neg q$

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Question 4. Write and check *tautology* of the logical statement. (10 points)

$$[(p \wedge q) \rightarrow r) \wedge (p \rightarrow q)] \rightarrow (p \rightarrow r)$$

Question 5. Verify and check *equivalent* of between the logical statements. (10 points)

$$(p \vee q) \rightarrow r \text{ and } \neg r \rightarrow (\neg p \wedge q)$$

Question 6. Given $U = \mathbb{R}$ (real numbers). Find the truth value of

$$\forall x [x^2 - 6x + 9 \geq 0] \leftrightarrow \exists x [\sqrt{x} = -x] \quad (10 \text{ points})$$

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Question 7. Find the negation of the quantified statement

$$\exists x \forall y \left[\frac{y^2}{x^2+1} - 2xy > (x^2 + y - 1)^2 \right]$$

where $U = \{0, 1, 4, 9, 16, \dots\}$. (10 points)

Question 8. Convert the following numbers to a number with base of ten (10 points)

8.1 $11111010001_2 =$

8.2 $FED_{16} =$

8.3 $(0.432)_5 =$

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Question 9. Convert the following numbers to a number with a given base (10 points)

9.1 $5604_8 =$ (base 12)

9.2 $32.311_4 =$ (base 16)

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Question 10. Simplify the following expressions (10 points)

10.1 $(2201_5 - 432_5) \times 14_5$

10.2 $(F12AB_{16} - EABC_{16}) \div 16_{16}$

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Question 11. Find the value of the expression with a given base (10 points)

11.1 $(2301031_4 + 630275_8) - 2FDA_{16}$ (base 8)

11.2 Solve the equation $(1411)_5 = (276)_x$

Assoc. Prof. Dr. Sanoë Koonprasert: Lecturer

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